

From One-Cluster SPC to Multicluster SPC: Distribution-Free Control Charts for Cleaning Validation Stability Analysis

1 d_ccssbu_l Function and d_ccssbu_l_d_selection_arl1 Function

- USL is used to form natural clusters in code but we use API to form natural clusters in the paper theoretical part.
- Functions `d_ccssbu_l()` can automatically detect 3A or 3B stages to generate control charts, algorithm's out in table.
- If we set `n_stage3A_events` to be a very large number, then the analysis will essentially be retrospective analysis.
- `d_ccssbu_l_d_selection_arl1()` used `d_ccssbu_l()` at `fit_q` and `fit_i` part.

1.1 d-CCSSBU-L Selection of d and ARL1 Results for Equipment A

- run the function `d_ccssbu_l_d_selection_arl1()`
- generate the csv file `d_CCSSBU_L_ARL1_Equipment_A_tbl_wide.csv` use `readr::write_csv()`
- generate Table 1 in the paper by the function `render_d_ccssbu_l_recommendation_tbl()` using argument `tbl = d_CCSSBU_L_ARL1_Equipment_A_recommendation`
- generate ARL₁ curve by `plot_d_ccssbu_l_arl1_curves()` and the curve picture name is `d-ccssbu-l-arl1-curves-equipment-a.jpeg`

Table 1: Recommended d values for the d-CCSSBU-L chart for Equipment A.

Residue	Recommended d	q	Decision case	Observed-behavior case	ARL1 case	A	U	W(d)	G(d)	Selection rule
DAR	3	3	Case 1	D1 intersection D2	U	{1, 2, 3}	{1, 2, 3}	1.0000	0.5659	U case: maximum d in U according to Rule 3
CAR	1	1	Case 2	D1 intersection D2	XO	{1}	Empty	36.0501	9.7164	XO case: minimum W over A; minimum G among candidates tied on W; maximum d if a tie remains according to Rule 3
Mic	1	1	Case 2	D1 intersection D2	XO	{1}	Empty	4.3094	0.8249	XO case: minimum W over A; minimum G among candidates tied on W; maximum d if a tie remains according to Rule 3

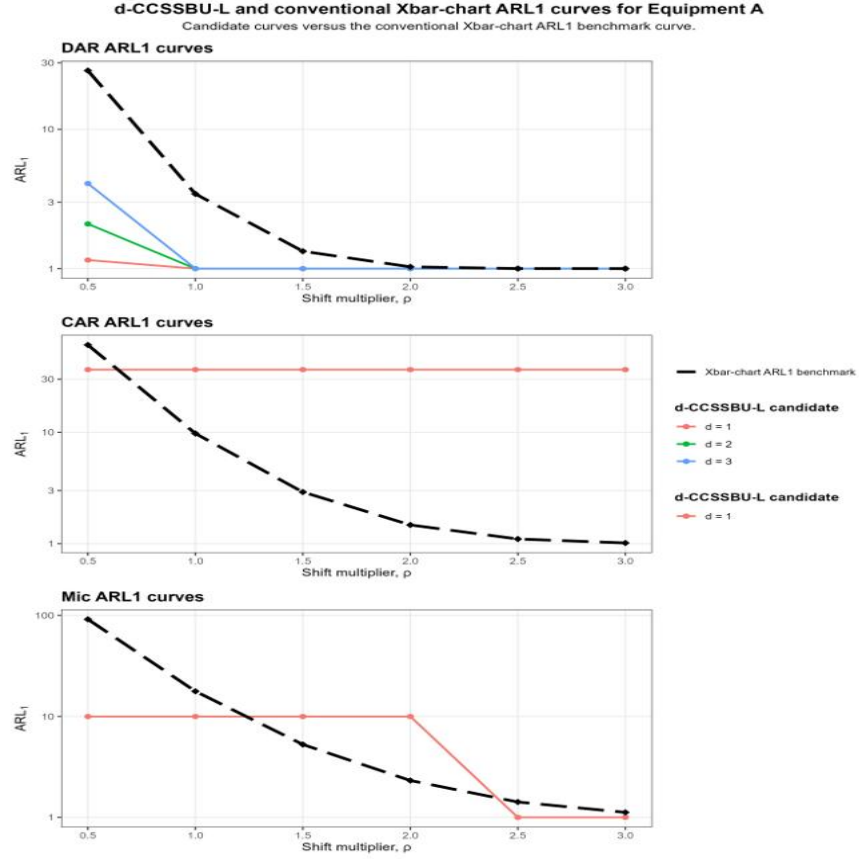


Figure 1: Candidate-specific d-CCSSBU-L ARL1 curves for Equipment A.

1.2 d-CCSSBU-L Selection of d and ARL1 Results for Equipment B

Repeat the procedure of above Equipment A

Table 2: Recommended d values for the d-CCSSBU-L chart for Equipment B.

Residue	Recommended d	q	Decision case	Observed-behavior case	ARL1 case	A	U	W(d)	G(d)	Selection rule
DAR	2	26	Case 1	D1 intersection D2	U	{1, 2, 3, 4, 5, 6, 7, 8, 9}	{2}	0.9999	0.4141	U case: maximum d in U according to Rule 3
CAR	2	2	Case 2	D1 intersection D2	XO	{2}	Empty	8.0175	2.2513	XO case: minimum W over A; minimum G among candidates tied on W; maximum d if a tie remains according to Rule 3
Mic	1	1	Case 2	D1 intersection D2	XO	{1}	Empty	26.0636	4.3581	XO case: minimum W over A; minimum G among candidates tied on W; maximum d if a tie remains according to Rule 3

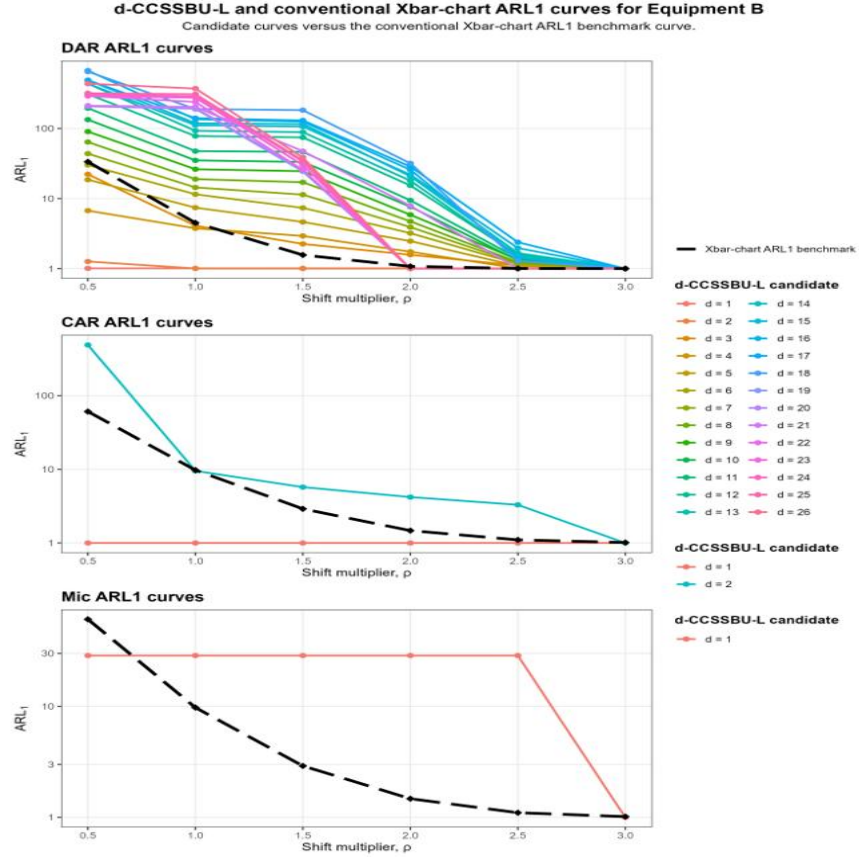


Figure 2: Candidate-specific d-CCSSBU-L ARL1 curves for Equipment B.

Above is for location chart and its ARL_1 curve and d selection. The following is for scale chart

2 d_ccssbu_s Function and d_ccssbu_s_d_selection_arl1 Function

- USL is used to form natural clusters in code but we use API to form natural clusters in the paper theoretical part.
- Functions `d_ccssbu_s()` can automatically detect 3A or 3B stages to generate control charts, algorithm's out in table.
- If we set `n_stage3A_events` to be a very large number, then the analysis will essentially be retrospective analysis.
- `d_ccssbu_s_d_selection_arl1()` used `d_ccssbu_s()` at `fit_q` and `fit_i` part.

2.1 d-CCSSBU-S Selection of d and ARL1 Results for Equipment A

- run the function `d_ccssbu_s_d_selection_arl1()`
- generate the csv file `d_CCSSBU_s_ARL1_Equipment_A_tbl_wide.csv` use `readr::write_csv()`
- generate Table 3 in the paper by the function `render_d_ccssbu_s_recommendation_tbl()` using argument `tbl = d_CCSSBU_S_ARL1_Equipment_A_recommendation`

- generate ARL_1 curve by `plot_d_ccssbu_s_arl1_curves()` and the curve picture name is `d-ccssbu-s-arl1-curves-equipment-a.jpeg`

Table 3: Recommended d values for the d -CCSSBU-S chart for Equipment A.

Residue	d	q	Case	Admissible Set	ARL Case	A	U	W	G	Rule
DAR	1	3	Case 2	$D1 \cap D2$	XO	1, 2	4.4881	1.5006		XO case: minimum W over A; minimum G among candidates tied on W; maximum d if a tie remains according to Rule 3
CAR	1	1	Case 2	$D1 \cap D2$	XO	1	4.5574	2.6012		XO case: minimum W over A; minimum G among candidates tied on W; maximum d if a tie remains according to Rule 3
Mic	1	1	Case 2	$D1 \cap D2$	XO	1	1.5395	0.1828		XO case: minimum W over A; minimum G among candidates tied on W; maximum d if a tie remains according to Rule 3

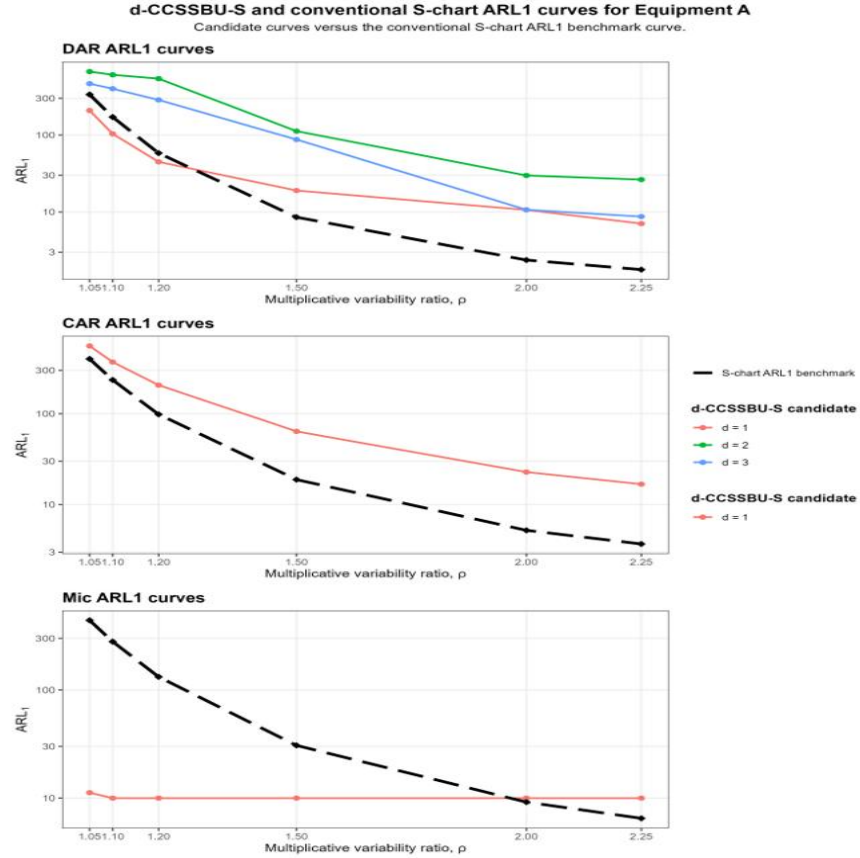


Figure 3: Candidate-specific d -CCSSBU-S ARL1 curves for Equipment A.

2.2 d -CCSSBU-S Selection of d and ARL_1 Results for Equipment B

Repeat as above Equipment A

Table 4: Recommended d values for the d -CCSSBU-S chart for Equipment B.

Residue	d	q	Case	Admissible Set	ARL Case	A	U	W	G	Rule
DAR	13	26	Case 4	D1	XO	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26	7.3323	3.0336		XO case: minimum W over A; minimum G among candidates tied on W; maximum d if a tie remains according to Rule 3
CAR	1	2	Case 2	$D1 \cap D2$	XO	1, 2	1.5221	1.1434		XO case: minimum W over A; minimum G among candidates tied on W; maximum d if a tie remains according to Rule 3
Mic	1	1	Case 2	$D1 \cap D2$	XO	1	35.9163	3.4234		XO case: minimum W over A; minimum G among candidates tied on W; maximum d if a tie remains according to Rule 3

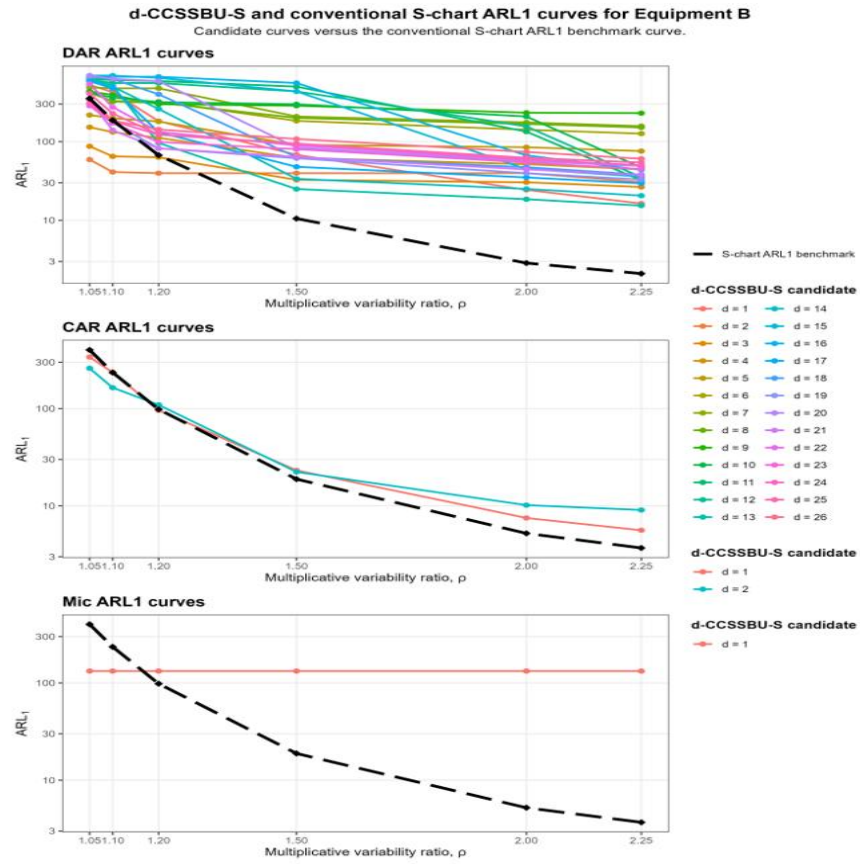
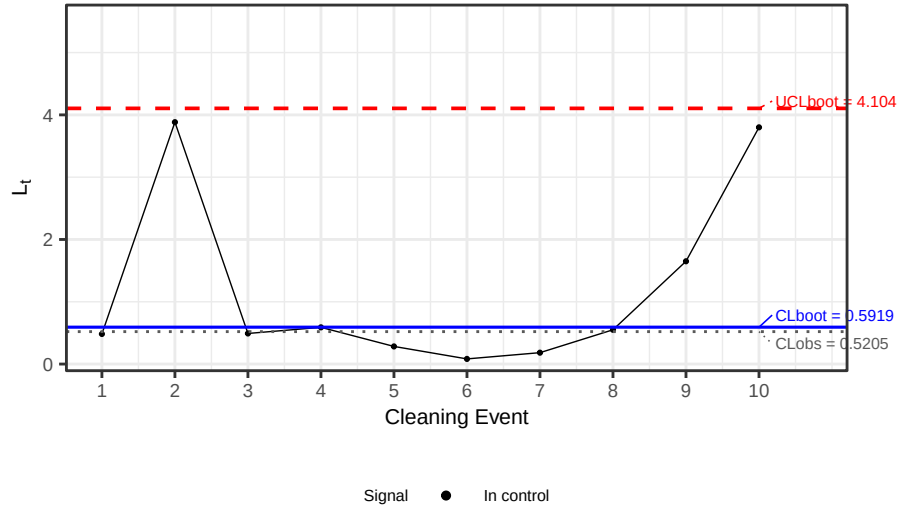


Figure 4: Candidate-specific d-CCSSBU-S ARL1 curves for Equipment B.

3 d-CCSSBU-L Charts for Recommended d Using Function $d_ccssbu_1()$

d-CCSSBU-L for DAR of Equipment A

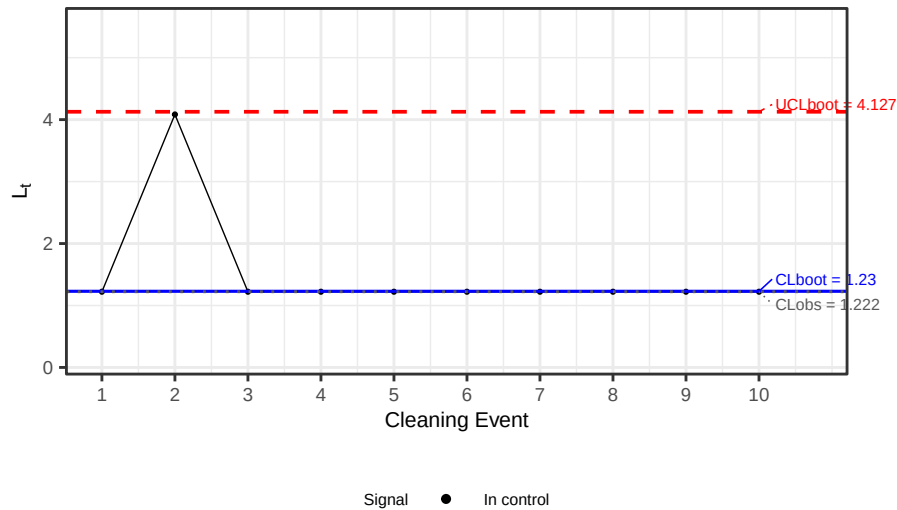
Case 2 | $d=3/3$ | $nCal=10$ | $UCLboot=4.104$ | $CLboot=0.5919$



Stage 3A OOC = 0; Stage 3B OOC = 0; $nCal = 10$; $d = 3/3$; Provisional limits

d-CCSSBU-L for CAR of Equipment A

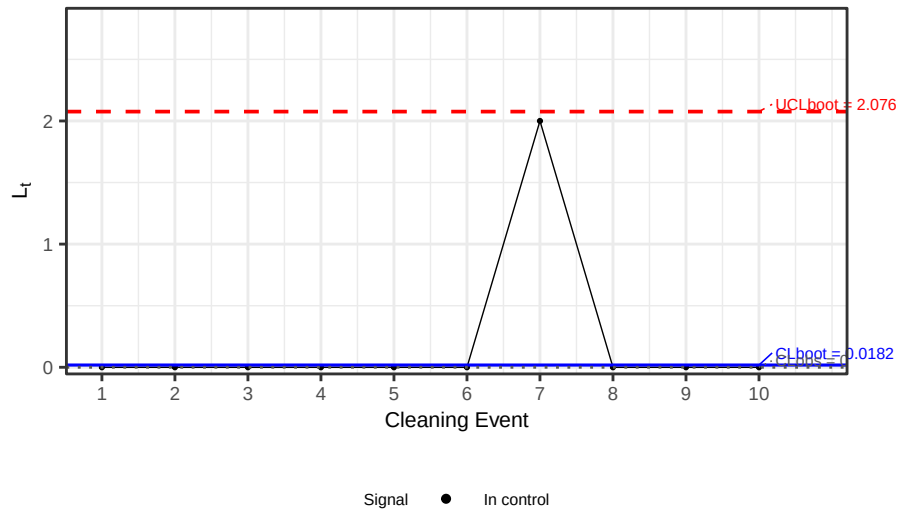
Case 1 | $d=1/1$ | $nCal=10$ | $UCLboot=4.127$ | $CLboot=1.23$



Stage 3A OOC = 0; Stage 3B OOC = 0; $nCal = 10$; $d = 1/1$; Finalized limits

d-CCSSBU-L for Mic of Equipment A

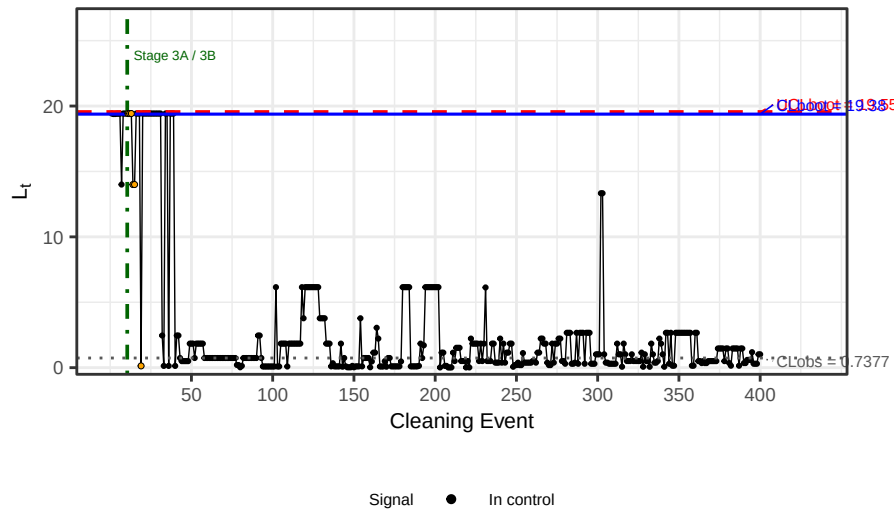
Case 1 | $d=1/1$ | $nCal=10$ | $UCL_{boot}=2.076$ | $CL_{boot}=0.0182$



Stage 3A OOC = 0; Stage 3B OOC = 0; $nCal = 10$; $d = 1/1$; Finalized limits

d-CCSSBU-L for DAR of Equipment B

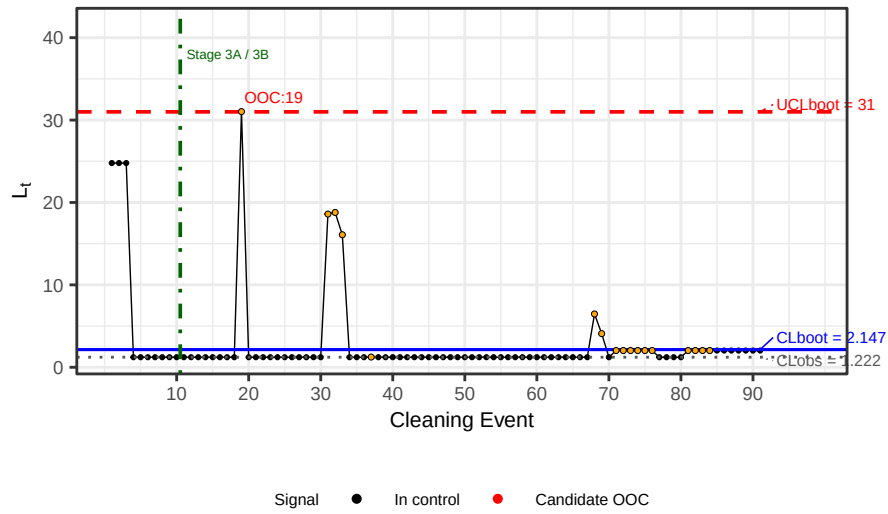
Case 5 | $d=2/26$ | $nCal=13$ | $UCL_{boot}=19.55$ | $CL_{boot}=19.38$



Stage 3A OOC = 0; Stage 3B OOC = 0; $nCal = 13$; $d = 2/26$; Provisional limits

d-CCSSBU-L for CAR of Equipment B

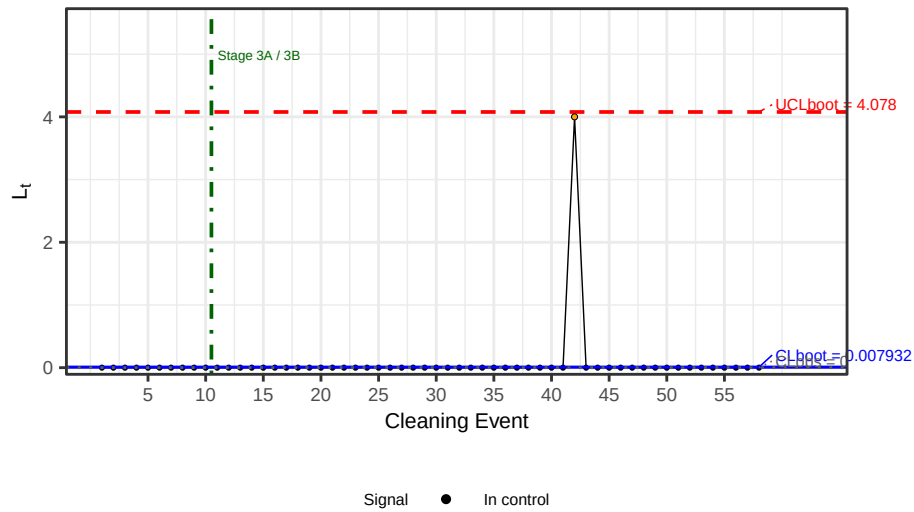
Case 4 | $d=2/2$ | $n_{Cal}=20$ | $UCL_{boot}=31$ | $CL_{boot}=2.147$



Stage 3A OOC = 0; Stage 3B OOC = 0; $n_{Cal} = 20$; $d = 2/2$; Finalized limits

d-CCSSBU-L for Mic of Equipment B

Case 4 | $d=1/1$ | $n_{Cal}=10$ | $UCL_{boot}=4.078$ | $CL_{boot}=0.007932$

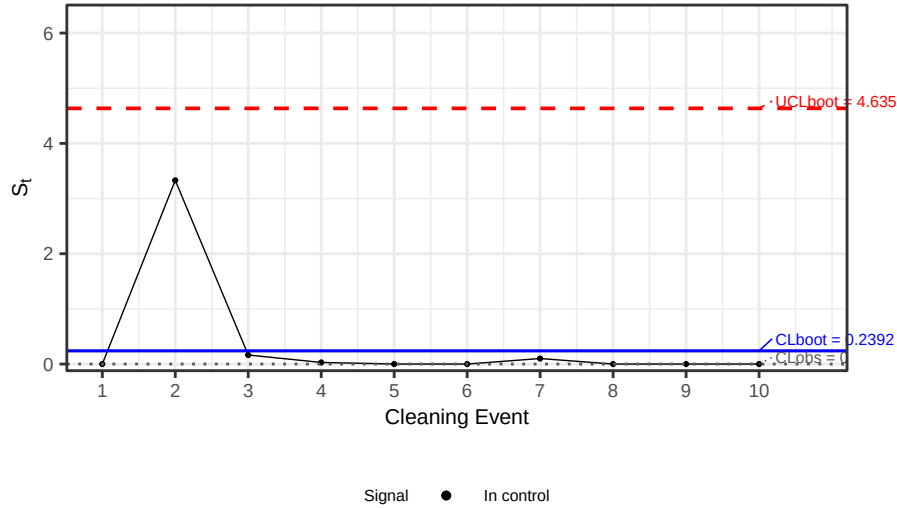


Stage 3A OOC = 0; Stage 3B OOC = 0; $n_{Cal} = 10$; $d = 1/1$; Finalized limits

4 d-CCSSBU-S Charts for Recommended d Using Function $d_ccssbu_s()$

d-CCSSBU-S for DAR of Equipment A

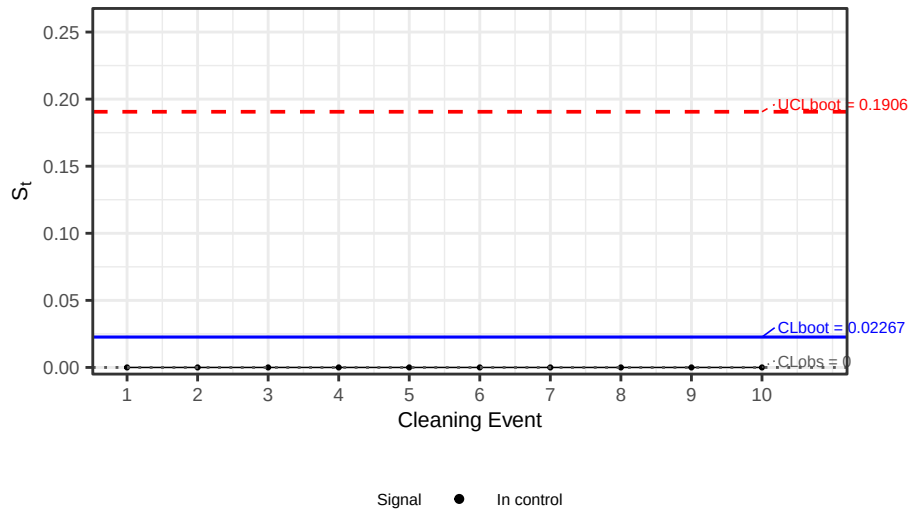
Case 2 | $d=1/3$ | $nCal=3$ | $UCLboot=4.635$ | $CLboot=0.2392$



Stage 3A OOC = 0; Stage 3B OOC = 0; $nCal = 3$; $d = 1/3$; Provisional limits

d-CCSSBU-S for CAR of Equipment A

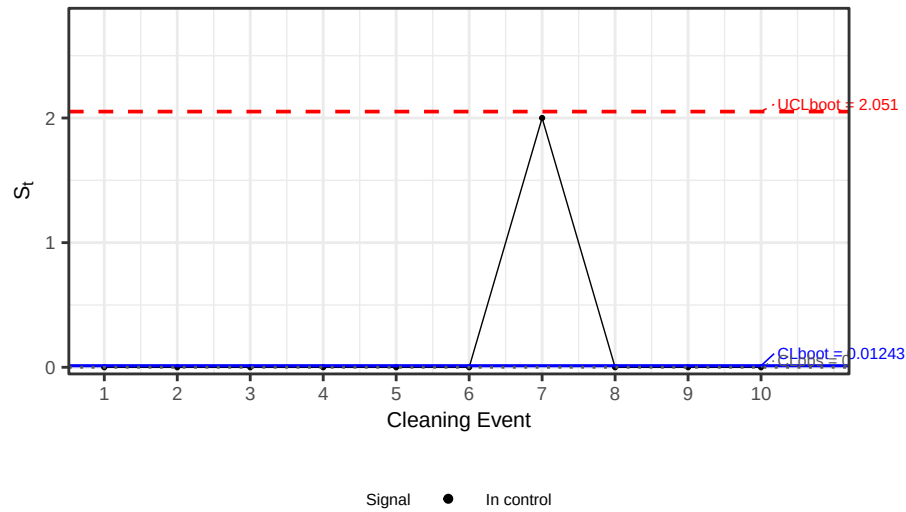
Case 1 | $d=1/1$ | $nCal=10$ | $UCLboot=0.1906$ | $CLboot=0.02267$



Stage 3A OOC = 0; Stage 3B OOC = 0; $nCal = 10$; $d = 1/1$; Finalized limits

d-CCSSBU-S for Mic of Equipment A

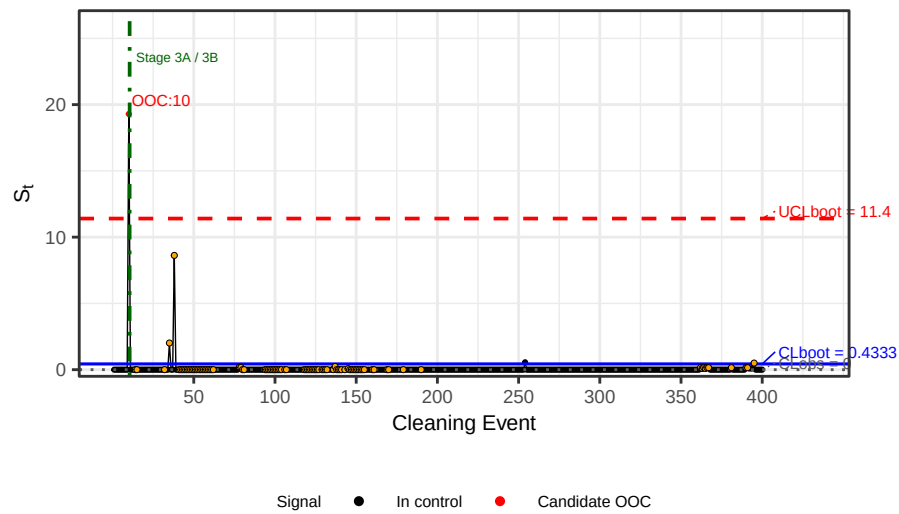
Case 1 | $d=1/1$ | $nCal=10$ | $UCL_{boot}=2.051$ | $CL_{boot}=0.01243$



Stage 3A OOC = 0; Stage 3B OOC = 0; $nCal = 10$; $d = 1/1$; Finalized limits

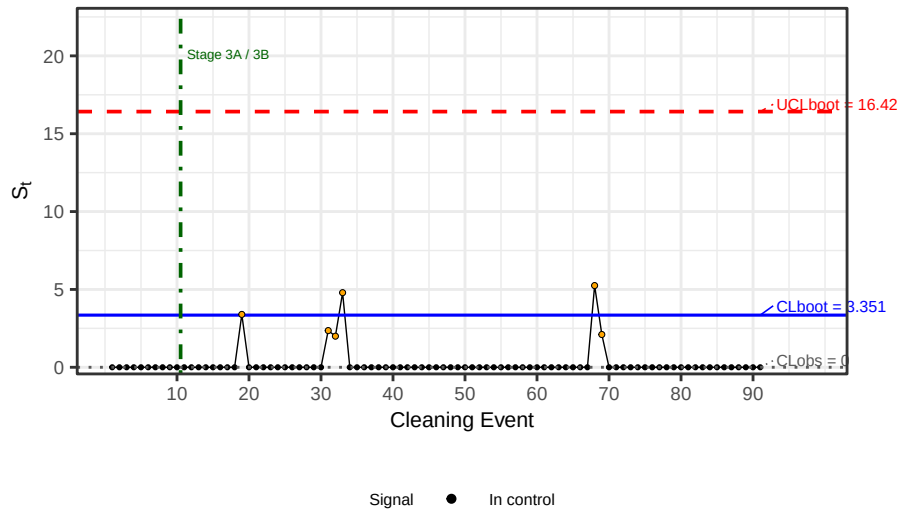
d-CCSSBU-S for DAR of Equipment B

Case 5 | $d=13/26$ | $nCal=101$ | $UCL_{boot}=11.4$ | $CL_{boot}=0.4333$



d-CCSSBU-S for CAR of Equipment B

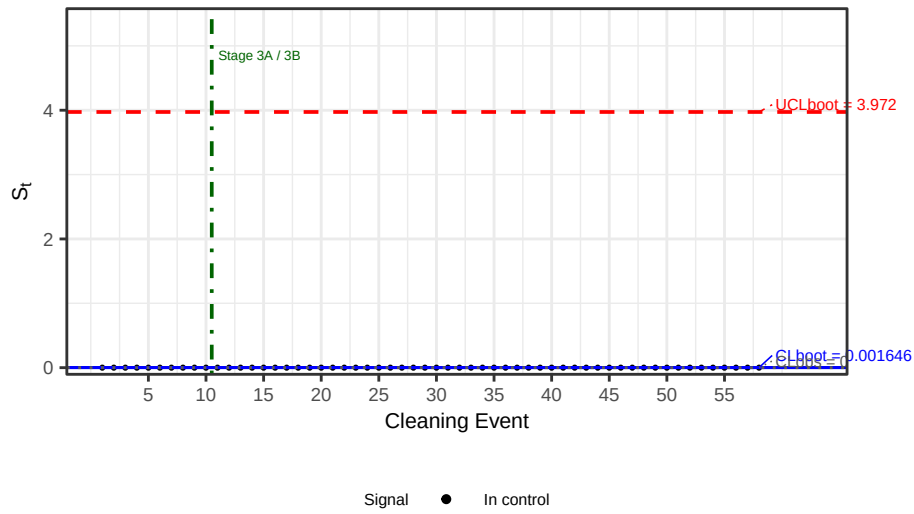
Case 4 | $d=1/2$ | $n_{Cal}=10$ | $UCL_{boot}=16.42$ | $CL_{boot}=3.351$



Stage 3A OOC = 0; Stage 3B OOC = 0; $n_{Cal} = 10$; $d = 1/2$; Finalized limits

d-CCSSBU-S for Mic of Equipment B

Case 3 | $d=1/1$ | $n_{Cal}=10$ | $UCL_{boot}=3.972$ | $CL_{boot}=0.001646$



Stage 3A OOC = 0; Stage 3B OOC = 0; $n_{Cal} = 10$; $d = 1/1$; Finalized limits

5.2 d-CCSSBU Charts for CAR of Equipment A and B Using Generated jpeg Files

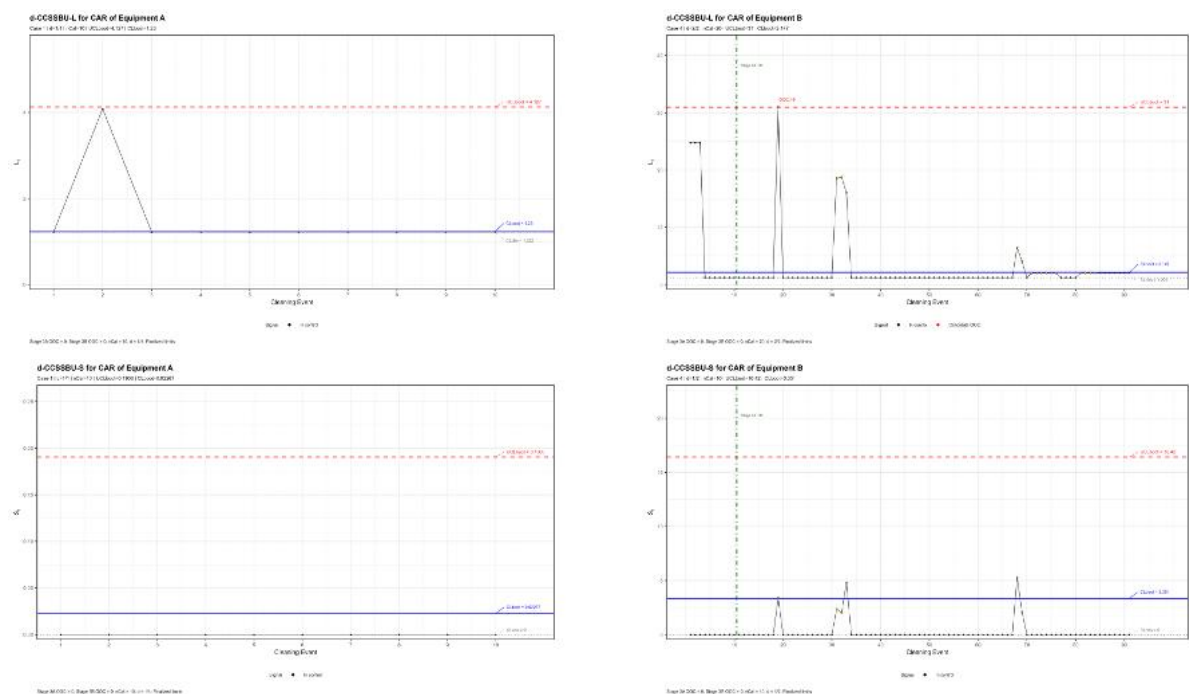


Figure 6: d-CCSSBU-L and d-CCSSBU-S charts for CAR of Equipment A and B.

5.3 d-CCSSBU Charts for Mic of Equipment A and B Using Generated jpeg Files

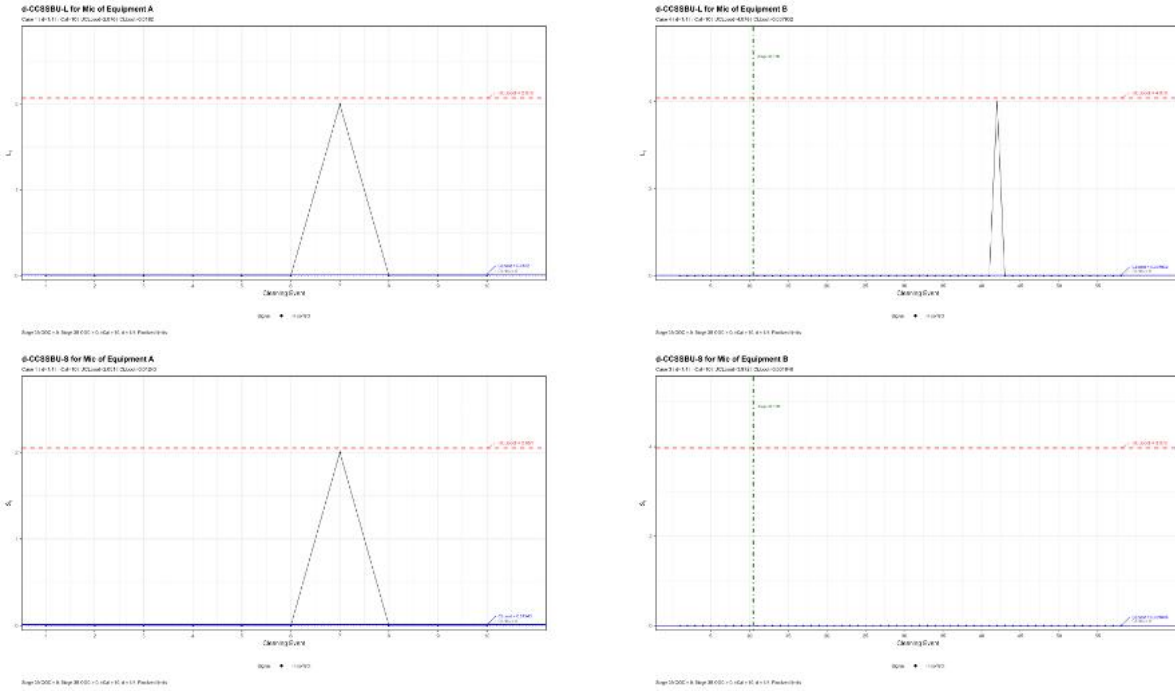


Figure 7: d-CCSSBU-L and d-CCSSBU-S charts for Mic of Equipment A and B.

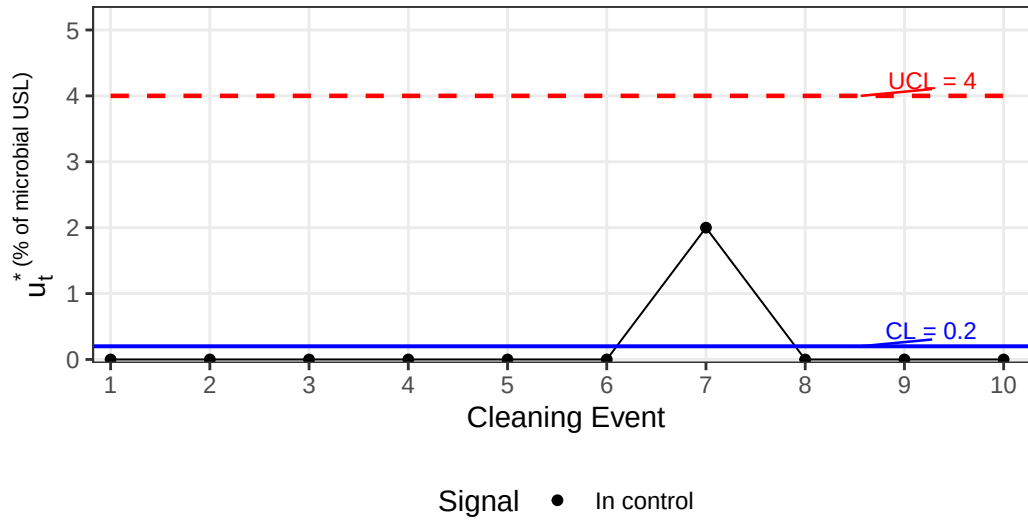
6 Poisson u Chart Illustration for Mic Data

The following is three-sigma Poisson u chart function `poisson_mic_u_chart()`.

The following is exact Poisson u chart function `poisson_mic_u_chart_exact()`.

Exact Poisson u Chart for Mic of Equipment

Stage 3A calibration | nCal events = 10 | exact one-sided Poisson UCL | ARL0 = 740 | CL = 0.2 | GOF



Exact UCL: $qpois(1 - 1/ARL0, p[t] * \bar{u}) / p[t]$; Stage 3A OOC = 0; Stage 3B OOC = 0. One-sided upper chart only.

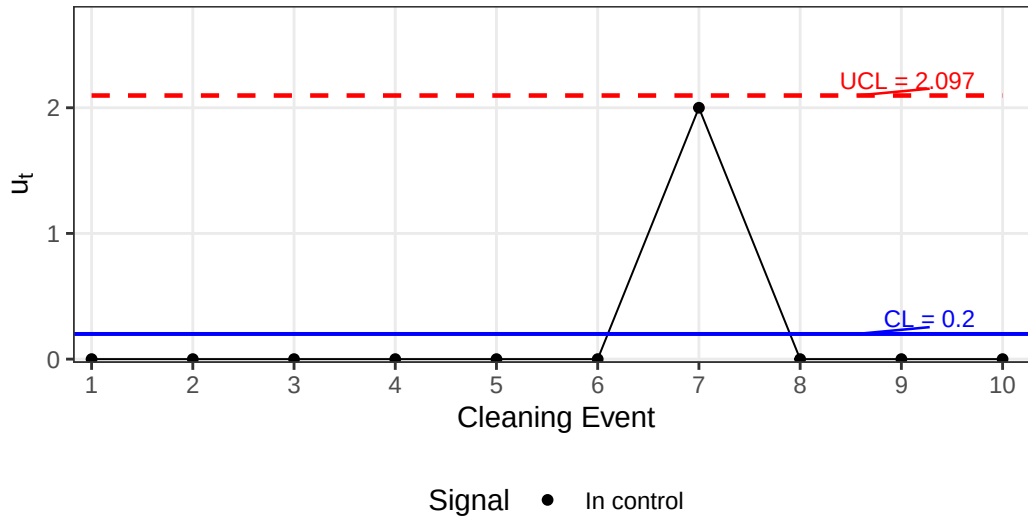
6.1 Three Sigma Poisson u Chart for Mic Data of Equipment A

Table 5: Mic data Poisson assessment and chart selection for Equipment A.

Equipment	Case	USL	GOF p	GOF pass	Dispersion	Overdispersion pass	Mic Poisson
Equipment A	Case 3	25	0.627	TRUE	1	TRUE	TRUE

Poisson u Chart for Mic of Equipment A

Initial Stage 3A analysis | nCal = 10 | CL = 0.2 | GOF p = 0.627 | dispersion ratio = 1



Stage 3A OOC = 0; Stage 3B OOC = 0; one-sided chart with no LCL.

6.2 Three Sigma Poisson u Chart and Exact Poisson u Chart Using Generated jpeg Files

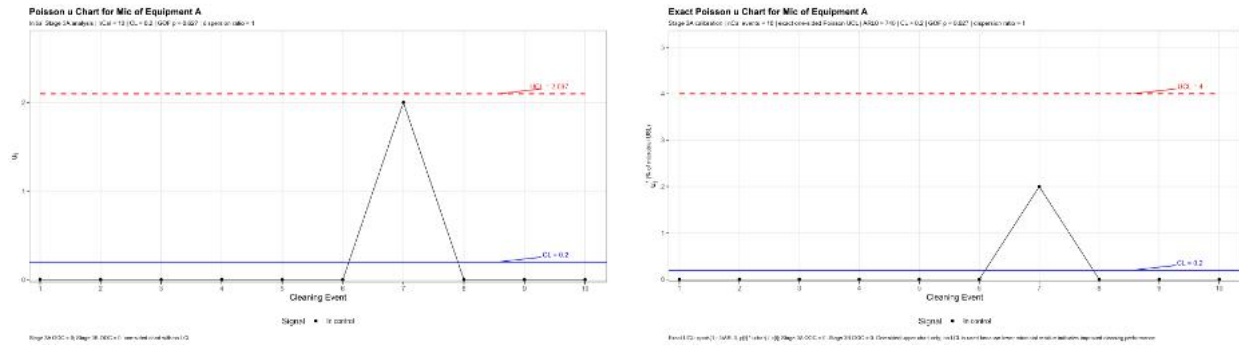


Figure 8: Comparison of conventional and exact Poisson u charts for Mic data from Equipment A.

6.3 Non-Poisson Mic Data of Equipment B

Table 6: Mic data Poisson assessment and chart selection for Equipment B.

Equipment	Case	USL	GOF p	GOF pass	Dispersion	Overdispersion pass	Mic Poisson
Equipment B	Case 3	25	0.0274	FALSE	1.6286	FALSE	FALSE